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PI Name:	Faeth, Gerard Michael Ph.D.		
Project Title:	Investigation of Laminar Jet Diffusion Flames in Microgravity: A Paradigm for Soot Processes in Turbulent Flames		
Division Name:	Physical Sciences		
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Comments:	Faeth, Gerard M.--DECEASED Jan 2005 (per N. Hall/GRC)		
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Task Description:	The structure and the soot formation and oxidation properties of laminar nonpremixed and premixed flames are being studied both experimentally and computationally; conditions where the intrusion of buoyancy is small are emphasized so that practical nonbuoyant flames can be approximated by exploiting classical laminar flamelet concepts. The investigation is divided into three phases, considering ground-based laminar premixed and nonpremixed flames and space-based laminar nonpremixed flames, respectively. This flight project is an experimental and theoretical investigation of the mechanisms of soot formation in laminar jet diffusion flames under conditions of low buoyancy. The flight experiment is in the Combustion Experiment Modules 1 and 2 (CM-1 and CM-2). In most flames of practical interest, soot radiation is the dominant mode of heat transport to combustor components and a dominant mechanism for flame spread and growth. The experimental results, combined with theoretical modeling, will either confirm or deny the applicability of the conserved scalar formalism to soot properties in diffusion flames. These results, when combined with those of ground-based low- and normal- gravity experiments will increase the current fundamental understanding of soot formation processes in both laminar and turbulent flames and increase our ability to model them. Consequently this will have a significant impact on our understanding of systems such as the spread of unwanted fires, design of jet engines and large scale boilers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
	Abstract. Work during this report period involved both flight- and ground-based experiments. Activities for the flight-based experiments consisted of reducing measurements of flame shapes and the distribution of soot properties within steady nonbuoyant soot-containing laminar jet diffusion flames during Space Shuttle Columbia flights STS-83, STS-94 and STS-107. These results were used to successfully evaluate a simplified model of the structure of these flames; subsequent application of this model suggested that soot properties within these flames should be universal functions of the degree of mixing of the fuel and the oxidant; these functions (called soot property state relationships) then provide a simplified and robust way of estimating soot properties in practical turbulent diffusion flames that otherwise would be computationally intractable for some time to come. This finding from the simplified model of flame structure was then confirmed by measurements of soot concentrations and continuum radiant emission from soot at wavelengths of 650 and 850 nm obtained from the observations of steady nonbuoyant laminar jet diffusion flames. These findings show that steady nonbuoyant laminar jet diffusion flames are the proper paradigm for soot processes in practical turbulent diffusion flames providing an experimentally tractable way to study soot processes in turbulent flames that will not be experimentally tractable for some time to come. The ground-based experiments involve detailed measurements of flame and soot properties in premixed and diffusion flame environments considering a variety of hydrocarbon fuels (CH4, C2H2, C2H4, C3H6, C3H8 and C6H6), gas temperatures of 1500-2350 K and pressures of 0.1-1.0 atm. In all cases, reaction rate properties were similar in premixed and diffusion flames and were independent of fuel type. Other conclusions of these measurements were as follows: primary soot particle surface growth rates were proportional to the product of the concentrations of C2H2 and H, and could be successfully correlated using existing Hydrogen-Abstraction-Carbon-Addition (HACA) mechanisms in the literature with steric factors on the order of unity, as expected; primary soot particle nucleation rates are crudely first-order with respect to the concentration of C2H2 and second-order with respect to the concentration of H, suggesting that nucleation is dominated by the formation of large PAH molecules via a HACA mechanism; and primary soot particle surface oxidation rates are dominated by the reaction with OH with a collision efficiency of 13%. Current work involves extending the range of the soot reaction property measurements to consider pressures of 1.0-10.0 atm and PAH as fuels in order to disclose any direct PAH reaction channels parallel to the HACA mechanism for soot surface growth. Flight-Based Experiments. The flight-based experimental apparatus (called the Laminar Soot Processes (LSP) Experiment) was successfully operated on the orbiting Space Shuttle Columbia during flights STS-83 in March 1997, STS-94 in July 1997 and STS-107 in January 2003. The test apparatus operated flawlessly and we were able to complete twenty-one (21) flame tests during flights STS-83 and STS-94, and twelve (12) flame tests during flight STS-107. No data was lost during flights STS-83 and STS-94, however, half the normally downlinked data and all of the sampling data was lost due to the unfortunate loss of the Space Shuttle Columbia and its crew during Space Shuttle Flight STS-107. Measured properties included flame images (lengths, diameters, and shapes), and distributions of soot properties in the flames (soot volume fractions, soot temperatures and soot radiative emission properties at two wavelengths based on imaging data and soot structure based on thermophoretic sampling and Transmission Electron Microscopy (TEM)). The test conditions considering all three flights were unusually broad and involved ethylene- and propane-fueled laminar jet diffusion flames burning in still air at pressures of 35-130 kPa, for burner diameters of 0.4-2.70 mm, and for burner exit Reynolds numbers of 46-1186. These conditions yielded steady nonbuoyant round laminar jet diffusion flames whose lengths ranged from soot-free (blue) flames to flames near the laminar smoke point. The first stage of data reduction involved evaluation of the classical analysis of the structure of steady nonbuoyant round laminar jet diffusion flames due to Spalding (1979), after empirically extending this analysis to account for luminosity due to the presence of soot within the flames. It was found that the extended Spalding (1979) analysis provided excellent predictions of the shape properties of the test flames, when radiative heat losses were small so that the intrusion of effects of thermal quenching were avoided. This analysis also shows that flame properties are identical functions of time along all soot paths emanating from the jet exit and subsequently convecting at the local velocity within the flame through the flame sheet to the ambient environment. In addition, the universality of flame properties as a function of time is also preserved even when jet exit velocities are varied. These characteristics imply that the soot properties of these flames are universal functions of the extent of mixing of the flow because soot particles convect through flames at local gas velocities, as well. As a result, soot in steady nonbuoyant round laminar jet diffusion flames should satisfy soot property state relationships in contrast to the behavior of soot properties in steady buoyant laminar jet diffusion flames where the intrusion of effects of buoyancy due to earth’s gravity prevents behavior that satisfies soot property state relationships. The existence of soot property state relationships was subsequently confirmed by direct measurements of soot volume fractions, and continuum radiant intensities at wavelengths of 650 and 850 nm, from the flight-based flames. Finally, the existence of soot property state relationships has four (4) important implications for practical applications, as follows: (1) it helps explain the surprising universality of soot properties within large hydrocarbon-fueled buoyant turbulent diffusion flames; (2) it helps explain why buoyant laminar jet diffusion flames do not exhibit soot property state relationships whereas these properties are observed for nonbuoyant laminar jet diffusion flames, which implies that only the latter are the proper laminar flame paradigm for the soot properties of practical turbulent diffusion flames; (3) it provides a tractable and robust way to develop methods of computational combustion for nonpremixed turbulent flames involving the combustion of hydrocarbons; and (4) it provides insight concerning the way that modifying the mixing properties of turbulent flames (e.g., by using air atomization for liquid-fueled propulsion

Task Progress:	systems) reduces the amount of soot formed and emitted from practical combustion processes. Finally, work during the next report period will involve extending the direct evaluation of the existence of soot property state relationships based on the imaging data available for laminar soot property flame experiments during Space Shuttle Columbia flights STS-83, STS-94 and STS-107. Ground-Based Experiments. Development of a fundamental understanding of the reaction properties of soot particles continued, based on detailed measurements of the structure of steady buoyant laminar jet diffusion flames. Earlier measurements involved both premixed and nonpremixed (diffusion) flame environments, a variety of hydrocarbon fuels (CH4, C2H2, C2H4, C3H6, C3H8 and C6H6), gas temperatures of 1500-2350 K, and pressures of 0.1-1.0 atm. These measurements were used to study primary soot particle surface growth rates, primary soot particle nucleation rates and primary soot particle early oxidation rates (before 70% of the maximum soot mass is oxidized after which the soot surface becomes porous and internal soot oxidation becomes a factor). These results showed that fundamental soot particle reaction properties were independent of flame type (premixed or diffusion), fuel type, temperature and pressure (in the range 0.1-1.0 atm). In particular, primary soot particle surface growth rates satisfied existing Hydrogen-Abstraction-Carbon-Addition (HACA) mechanisms due to Kazakov et al. (1995) and Colket and Hall (1994), with unknown steric factors in these mechanisms having values on the order of unity, as expected. In addition, primary soot particle nucleation rates were correlated effectively by assuming that the rate of development of large PAH molecules by the Hydrogen-Abstraction-Carbon-Addition (HACA) mechanism controlled primary soot particle nucleation rates having first and second molecular orders with respect to the concentrations of acetylene and H, respectively. Finally, early primary soot particle surface oxidation rates for all the flames were found to satisfy the OH/O2 surface oxidation mechanism that Neoh et al. (1980) established based on measurements in the post-flame region of laminar premixed flames with a collision efficiency of OH for early soot oxidation of 13% and a relatively weak direct O2 reaction pathway based on the classical results of Nagle and Strickland-Constable (1962). Given these results concerning primary soot particle surface growth rates, primary soot particle nucleation rates and early primary soot particle oxidation rates in laminar premixed and diffusion flames, for various hydrocarbon fuels (CH4, C2H2, C2H4, C3H6, C3H8 and C6H6), gas temperatures of 1500-2350 K, and pressures of 0.1-1.0 atm, information needed to simulate soot reaction properties in flame environments is generally in hand. There are two areas, however, where additional information is needed, as follows: (1) the potential effect of polycyclic aromatic hydrocarbon (PAH) fuels, which could enhance primary soot particle surface growth and nucleation rates compared to the flame environments considered thus far; and (2) effects of pressure on soot particle reaction rates at pressures greater than atmospheric pressure that are relevant to numerous practical combustion applications. In order to fill these gaps in the literature detailed measurements of flame structure and soot properties are being undertaken to consider a typical PAH fuel (toluene) burning in an air coflow at atmospheric pressure, and acetylene fueled flames burning in an oxidizing coflow at pressures of 1.0-10.0 atm. The specific test conditions involve a total of seven (7) steady round buoyant laminar jet diffusion flames, with the fuel jets directed vertically upward and soot and flame structure measurements limited to various positions along the axes of the flame, similar to past work. The measurements are also similar to past work: soot volume fractions using deconvoluted laser absorption, primary soot particle diameters using thermophoretic sampling and analysis by Transmission Electron Microscopy (TEM) and High Resolution Transmission Electron Microscopy (HRTEM), flow (soot) temperatures using deconvoluted multiline emission, flow velocities using laser velocimetry, concentrations of major stable gas species by isokinetic sampling and analysis using gas chromatography, and concentrations of some radial species (H, OH and O) using the deconvoluted Li/LiOH atomic absorption technique. Observations for the seven new flames will be combined with past observations of the twenty-two (22) earlier flames in order to study effects of local flame properties on soot surface reaction properties. Work completed thus far has involved establishing test conditions for the three toluene-fueled flames at atmospheric pressure and the four (4) acetylene-fueled flames at pressures of 1.0-10.0 atm. Complete measurements for each series of flames, along with data reduction and analysis of the measurements, takes roughly one man year. We are well along with the measurements of both series of flames and will complete this phase of the investigation during the next report period. Understanding the mechanisms of soot reactions in flame environments on earth has significant relevance to society. Several specific effects of soot are important to both public health and combustion technologies, as follows: pollutant soot emitted from combustion processes causes more deaths than any other pollutant, comprising roughly 60,000 premature deaths each year in the US alone; carbon monoxide emissions intrinsically associated with soot emissions from flames cause 80 percent of the fatalities of unwanted fires, comprising roughly 4,000 deaths each year in the US alone; continuum radiation from soot is mainly responsible for the growth and spread of unwanted fires, causing 5,000 deaths each year in the US alone; particulate soot emitted from combustion processes has recently been identified as a major contributor to global warming; particulate soot emitted from combustion processes has recently been shown to cause melting of ice and snow in arctic regions by settling on their surfaces and increasing their absorption of solar radiation; and successful development of computational combustion is improbable without an understanding of soot reaction properties in flame environments. These observations amply motivate study of soot reaction properties in flames.

Bibliography Type:	Description: (Last Updated: 10/19/2009)
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